

Ejercicios cuaderno transformada inversa de Laplace:

Primero Vamos a encontrar la transformada inversa de Laplace de:

$$X(s) = \frac{2s^2 + 14s + 124}{s^3 + 8s^2 + 46s + 68}$$

$$\operatorname{Re}\{s\} > -2$$

Factorizando el denominador (polinomio cúbico)

$$s^3 + 8s^2 + 46s + 68 \rightarrow \boxed{s = -2} \quad (-2)^3 + 8(-2)^2 + 46(-2) + 68 = 0$$

Dividiendo el polinomio entre el factor $(s+2) \rightarrow s+2=0$
 $s = -2$

$$s^3 + 8s^2 + 46s + 68 = -(s+2)(s^2 + 6s + 34)$$

Descomponiendo en fracciones parciales

$$\frac{2s^2 + 14s + 124}{(s+2)(s^2 + 6s + 34)} = \frac{A}{s+2} + \frac{Bs+C}{s^2 + 6s + 34}$$

$$2s^2 + 14s + 124 = A(s^2 + 6s + 34) + (Bs+C)(s+2)$$

$$A(s^2 + 6s + 34) = As^2 + 6As + 34A$$

$$(Bs+C)(s+2) = Bs^2 + 2Bs + Cs + 2C = Bs^2 + (2B+C)s + 2C$$

$$2s^2 + 14s + 124 = (A+B)s^2 + (6A+2B+C)s + (34A+2C)$$

Iguando coeficientes

$$s^2: A+B = 2$$

$$s: 6A+2B+C = 14$$

$$\text{const: } 34A+2C = 124$$

Resolviendo el sistema:

$$A+B = 2 \rightarrow B = 2-A \quad (1)$$

$$6A+2B+C = 14 \quad (2)$$

$$34A+2C = 124 \quad (3)$$

sustituyendo (1) en (2)

$$6A+2(2-A)+C = 14 \rightarrow 6A+4-2A+C = 14 \rightarrow 4A+C = 10 \quad (4)$$

$$C = 10-4A$$

sustituyendo (4) en (3)

$$34A+2C = 124 \rightarrow 34A+2(10-4A) = 124 \rightarrow 34A+20-8A = 124$$

$$26A = 104 \rightarrow A = \frac{104}{26} \rightarrow \boxed{A = 4}$$

Entonces, $A = 4$; $B = 2 - A \rightarrow B = -2$; $C = 10 - 4A = 10 - 16 = -6$

$C = 10 - 4(4) \rightarrow C = -6$

Reescribiendo la expresión

$$X(s) = \frac{4}{s+2} + \frac{-2s-6}{s^2+6s+34}$$

Aplicando transformada de Laplace inversa

$$\mathcal{L}^{-1}\left\{\frac{4}{s+2}\right\} = 4e^{-2t}$$

$$\mathcal{L}^{-1}\left\{\frac{-2s-6}{s^2+6s+34}\right\}$$

$$s^2+6s+34 = (s+3)^2 + 125$$

$$-2s-6 = -2(s+3)$$

Entonces

$$\mathcal{L}^{-1}\left\{\frac{-2s-6}{s^2+6s+34}\right\} = -2e^{-3t}\cos(5t)$$

Resultado final:

$$x(t) = \mathcal{L}^{-1}\{X(s)\} = 4e^{-2t} - 2e^{-3t}\cos(5t)$$

Ahora vamos a (aplicar) obtener la transformada inversa de Laplace manualmente de:

$$X(s) = \frac{1}{(s+1)(s+2)^2} \quad \Re\{s\} \geq -1$$

Descomponiendo en fracciones Parciales tenemos:

$$\frac{1}{(s+1)(s+2)^2} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{(s+2)^2}$$

Multiplicando ambos lados por $(s+1)(s+2)^2$

$$\frac{(s+1)(s+2)^2}{(s+1)(s+2)^2} = \frac{A(s+1)(s+2)^2}{s+1} + \frac{B(s+1)(s+2)^2}{s+2} + \frac{C(s+1)(s+2)^2}{(s+2)^2}$$

$$1 = A(s+2)^2 + B(s+1)(s+2) + C(s+1)$$

Expandiendo cada termino

$$A(s+2)^2 = A(s^2 + 4s + 4)$$

$$B(s+1)(s+2) = B(s^2 + 3s + 2)$$

$$C(s+1) = C(s+1)$$

$$1 = A(s^2 + 4s + 4) + B(s^2 + 3s + 2) + C(s+1)$$

Agrupando potencias de s y igualando coeficientes

$$1 = (A+B)s^2 + (4A+3B+C)s + (4A+2B+C)$$

$$A+B=0 \quad (1)$$

$$4A+3B+C=0 \quad (2)$$

$$4A+2B+C=1 \quad (3)$$

Resolviendo el sistema

$$A+B=0 \rightarrow B=-A$$

$$4A+3(-A)+C=0 \rightarrow A+C=0; \quad C=-A$$

$$4A+2(-A)+(-A)=1 \rightarrow 4A-2A-A=1; \quad \boxed{A=-1}$$

$$B=-(-1) \rightarrow \boxed{B=1} \quad C=A \rightarrow \boxed{C=-1}$$

$$\frac{1}{(s+1)(s+2)^2} = \frac{-1}{s+1} + \frac{1}{s+2} - \frac{1}{(s+2)^2}$$

Luego, aplicando propiedades de la transformada inversa de Laplace.

$$\mathcal{L}^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at} u(t), \quad \mathcal{L}^{-1}\left\{\frac{1}{(s+a)^2}\right\} = te^{-at} u(t)$$

Entonces: $x(t) = -e^{-t} + e^{-2t} + te^{-2t}, \quad t \geq 0$

$$x(t) = \mathcal{L}^{-1}\{x(s)\} = (-e^{-t} + e^{-2t} + te^{-2t}) u(t)$$